BRIDGING 2D EFFICIENCY AND 3D CONTEXT: A MEMORY-GUIDED FRAMEWORK FOR KNEE MRI MULTI-LABEL CLASSIFICATION

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ABSTRACT

Magnetic Resonance Imaging (MRI) plays a crucial role in detecting and classifying anatomical structures. However, automated analysis of medical magnetic resonance imaging remains challenging due to the many anatomical structures (e.g., the ACL and meniscus in the knee) and surrounding soft tissues that have very similar signal intensities, making their boundaries difficult to classify clearly. In this paper, we present a lightweight, end-to-end framework that bridges 2D and 3D analysis. By using a multi-scale center cropping strategy to capture anatomical details at differing resolutions and employing task-aware memory with depth-aware attention, our method automatically classifies the most diagnostic angle for each injury while maintaining high computational efficiency for clinical deployment. In evaluations on the MRNet dataset for the knee, our framework demonstrates excellent performance in labeling volumetric sequences, while also providing a mechanism to highlight the relevant anatomical features that contribute to diagnosis, thereby bridging the gap between automated scores and clinical reasoning. By combining multi-scale inputs with precise, context-aware feature encoding, our framework provides a robust, lightweight, and interpretable solution for diagnosing complex knee injuries.

Index Terms— Explainable AI, Medical Image Analysis, Single-view, Memory Network, Lightweight

1. INTRODUCTION

Magnetic Resonance Imaging (MRI) has long been recognized as a robust tool in modern medical diagnosis, offering excellent visualization of soft tissues, internal organs, and complex anatomical structures [1] [2]. Its widespread use is due to the ability to provide high-resolution, multi-view slices, which enable the classification of exact injury details that cannot be visible with other methods. However, looking at an MRI slice is a complex and time-consuming process [1]. Clinicians face significant challenges in classifying overlapping slices and in different views (sagittal, coronal, and axial planes), while distinguishing between pathological findings and normal physiological variations. Due to this complexity, compounded by the similar contrast between the injury and surrounding tissues, it frequently leads to misdiagnosis and differing viewpoints among clinicians. Consequently, there is a growing imperative for the advancement of automated Computer-Aided Diagnosis (CAD) systems in modern healthcare [3]. Despite the growing field of deep learning in medical imaging, developing an automated framework that is both robust and efficient is still a significant challenge. Existing methods often face a trade-off between computational efficiency and volumetric understanding. Computing in each slice independently, using 2D models such as Convolutional Neural Network (CNNs), is lightweight and data-efficient; however, they fail to capture the continuous structural dependencies necessary to identify diffuse lesions that span multiple slices. Conversely, while 3D models capture global context, they are computationally expensive and prone to overfitting due to the limited availability of annotated medical data [5] [6]. Additionally, other methods using multi-view integration techniques often utilize an inflexible concatenation approach, which implicitly assumes that every anatomical structure is equally informative in every view. However, in practice, pathologies are not always visible across all views. Often, one specific view is optimal for diagnosing a particular condition, with the others providing supplementary context. [1] [3] [4] [5] [6].

To overcome this limitation, we propose an end-to-end anatomy-aware framework that pursues two overarching objectives: **(i) efficient volumetric reasoning**, bridging the computational efficiency of 2D processing with the volumetric context of 3D understanding; **(ii) inherent interpretability**, which reduces the black-box nature of the model by explicitly indicating the anatomical regions that contribute to the diagnostic decision. **Our key contributions are:**

- **Multi-Scale Context Integration** (objective: *zooming to the anatomy center*). Instead of static resizing, we employ a multi-scale center cropping strategy that emulates the radiologist's workflow of focusing on the anatomical center. This approach ensures the model captures essentials structural alignment and fine-grained textural anomalies.
- **Hybrid Depth-Aware Encoding** (objective: *efficient volumetric reasoning*). To overcome the limitations of 2D

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Our code is available at GitHub.

slice-wise processing and heavy 3D networks, we propose a lightweight backbone enhanced with Depth-Aware Attention and CBAM [7]. Unlike rigid max-pooling, our learnable attention mechanism dynamically weights the informative slices within the MRI volume while preserving spatial granularity.

- **Task-Aware Memory** (objective: *pathology-specific feature enhancement*). We propose a method that stores different feature representations for each type of injury, acting as a specialized knowledge base for individual pathologies [8]. By using this memory component, the model can effectively separate co-occurring diseases, enabling it to distinguish which pathological patterns correspond to specific injuries.

Experimental results on the MRNet dataset [1] demonstrate that our framework achieves high AUC scores across individual views settings . The model maintains a lightweight architecture while providing transparent visualizations of pathological regions, improving clinical reliability.

2. RELATED WORK

The slice-based 2D Convolutional Neural Network (CNN) framework has become the main approach in automated knee MRI analysis. Most methods employing this architecture use standard 2D backbones such as Resnet to extract features from individual MRI slices independently, treating the input volume as a sequence of images. Subsequently, pooling mechanisms or recurrent networks are employed to aggregate slice-wise representations, generating a final diagnostic prediction based on the scan's global context.

However, a key limitation of these approaches lies in their reliance on global features extracted from the entire image slice. Such a global representation often overlooks fine-grained region-specific details, making it difficult for the model to capture subtle pathological changes within individual anatomical structures and increasing sensitivity to background noise. Moreover, extending these methods to full 3D models typically leads to a dramatic increase in computational cost. Drawing inspiration from natural language processing, a memory-augmented network has shown to be an effective paradigm to store dataset-level contextual information and retrieve feature prototypes, showing great potential in disentangling overlapping clinical patterns without the overhead of 3D Convolutions. For example, Huang et al. [8] successfully applied a trainable memory matrix to guide feature extraction in knee MRI, significantly enhancing the model ability's to focus on special features and eliminate noise.

Inspired by the advances, we propose a lightweight framework that improves feature perception by introducing a structured and semantically meaningful representation. Specifically, we integrate unsupervised joint localization with a novel task-aware memory mechanism. While previous memory modules focus on dataset-level or plane-level context, our task-aware disentangles for specific pathologies (e.g., ACL vs. Meniscus), acting as a knowledge base to separate co-occurring diseases efficiently.

3. METHOD

3.1. Overview

Our framework processes the Sagittal view MRI sequence to diagnose multiple knee pathologies (ACL tear and Meniscus tear). The pipeline consists of three main stages: (1) **Feature Extraction**, which encodes the MRI slices into spatial representations, (2) **Task-Aware Memory Module**, which refines the features of each pathology by using anatomical attention and historical prototypes, and (3) **Classification**, which gives the final probabilities for ACL and Meniscus tear.

3.2. Feature Extraction with Multi-Scale Context

To capture the global and local structure of the MRI images, we employed a multi-scale cropping strategy for each slice. The input MRI volume is processed as global, mid-level, and close-up views corresponding to crop ratios **1.0**, **0.65**, and **0.55**, respectively. All views are resized to 224×224, which are the input of the backbone.

We first process the input MRI sequence, represented as a volume of dimensions $R^D \times 3 \times 224 \times 224$ (where D represents the depth of the sequence), in a ResNet-18 backbone [9]. We expand the channel dimension to 3 to construct a multi-scale representation. Specifically, the three channels correspond to distinct spatial scales extracted from each slice: a global view, a mid-level zoom, and a localized crop. This composite representation enables the network to capture multi-resolution features, ranging from global anatomical context to fine-grained lesions. At the same time, it matches the 3-channel input format of the pre-trained ResNet backbone, facilitating effective transfer learning. To effectively mitigate the domain shift between standard natural images and medical scans, this backbone is initialized with weights pre-trained on RadImageNet [10] utilizing Multi-Label Supervised Contrastive Learning [11]. In a standard ResNet architecture, aggressive downsampling can remove pathological features that are important for knee diagnosis. To overcome this issue, we modify the final residual block by removing its stride. This change preserves a higher spatial resolution and produces a feature volume of size $R^D \times 512 \times 14 \times 14$, which helps to retain the details of the small lesion. The resulting feature volume is then used as the input to our attention-based refinement stages.

We employ the Convolutional Block Attention Module (CBAM) [7]to mitigate the low contrast between soft tissues and surrounding fluids. CBAM [7]operates independently on each of the D slices and sequentially generates attention maps

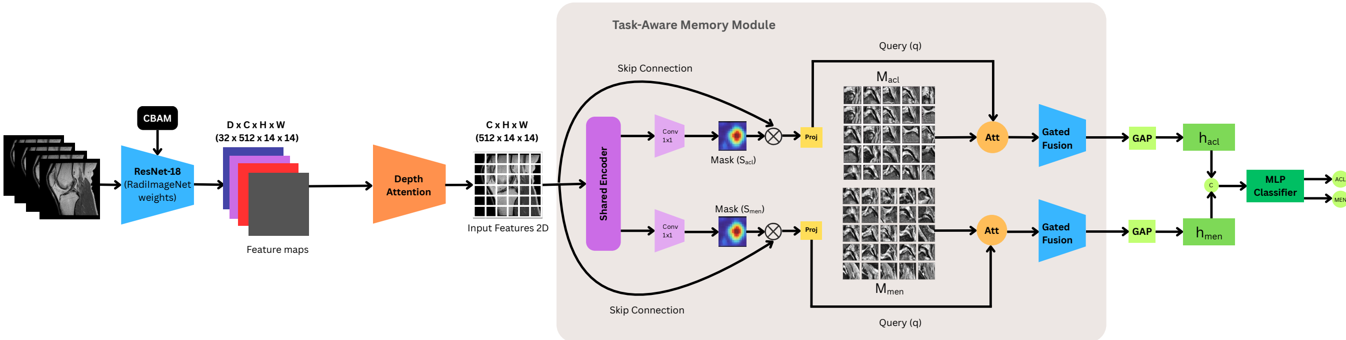


Fig. 1: **Architecture of the proposed SingleViewMemoryNet pipeline.** First, the input MRI slices are processed by a RadImageNet-pretrained ResNet-18 backbone integrated with a CBAM module to extract multi-slice feature maps. Next, the Depth Attention module evaluates slice importance to compress the $D \times C \times H \times W$ volume into a 2D feature representation ($C \times H \times W$). These features then enter the Task-Aware Memory Module, where a shared encoder generates task-specific spatial masks (S_{acl} and S_{men}) via skip connections to isolate relevant anatomical regions. The masked features act as queries (q) to retrieve explicit diagnostic patterns from task-specific memory banks (M_{acl} and M_{men}). Finally, following a Gated Fusion and Global Average Pooling (GAP), the refined task embeddings (h_{acl} and h_{men}) are concatenated and passed through an MLP classifier for simultaneous ACL and Meniscus tear prediction.

to better filter noise within each slice. First, its channel attention sub-module utilizes both average and max pooling followed by a shared Multi-Layer Perceptron (MLP) to automatically highlight the most informative feature channels. Next, the spatial attention sub-module pools the features along the channel dimension and applies a 7×7 convolution. This step highlights the structural boundaries of critical anatomical regions within each slice while assigning low weights to less informative background areas (e.g., irrelevant fluid signals). The output of this stage remains a refined feature tensor of size $R^D \times 512 \times 14 \times 14$.

Subsequently, to capture inter-slice dependencies across each feature channel, we permute the refined feature volume into $\mathcal{X} \in R^{C \times D \times H \times W}$ (where $C = 512, H = 14, W = 14$) and fed into a Depth-Aware Attention layer. Since knee pathologies (e.g., meniscus tears) are often sparse and localized in only a few slices, standard pooling methods (such as average or max pooling) can dilute or discard these subtle diagnostic signals with background noise. In contrast, our module selectively evaluates and highlights slices that contain pathological evidence.

The module first applies Global Average Pooling (GAP) to summarize each slice while discarding spatial details. In convolutional feature maps, the C channels act as pattern detectors. When a pathology is present, it induces high activations in localized regions of the $H \times W$ grid for specific channels. By averaging over the spatial gird, GAP captures this behavior: channels that respond to lesions produce higher average values. As a result, the spatial dimensions are reduced from $H \times W$ to 1×1, resulting in a descriptor $G \in R^{C \times D \times 1 \times 1}$ that summarizes the responses of the channels for each slice.

The network then estimates the diagnostic importance of the signals summarized in G using a 3D information bottleneck. To maintain computational efficiency and prevent noisy slices from corrupting informative ones, we deliberately avoid mixing features across the depth dimension at this stage. Instead, G is passed through a channel reduction layer parameterized by weights $\mathbf{W}_1 \in R^{\frac{C}{r} \times C}$, followed by a ReLU activation (δ) [12].

The features are then expanded back to the original dimension using an expansion layer with weights $\mathbf{W}_2 \in R^{C \times \frac{C}{r}}$. In practice, $\mathbf{W}_1$ and $\mathbf{W}_2$ are strictly implemented as $1 \times 1 \times 1$ 3D convolutions. Rather than full cross-slice mixing, this design acts as a channel-wise recalibration per slice. It operates strictly along the channel dimension while sharing weights across the depth, where r denotes the reduction ratio ($r = 8$). This independent squeeze-and-expand architecture serves two purposes. First, by compressing the channel dimension from C to $\frac{C}{r}$, we introduce an information bottleneck that encourages the network to learn compact representations, due to the limited capacity of the intermediate layer, the model is encouraged to find the most informative features. The reduction weights W_1, optimized through backpropagation, allow the network to form linear combinations of input features, suppressing redundant signals (e.g., background or healthy tissue) while preserving salient pathological patterns in the lower-dimensional space. The expansion layer (W_2) then projects the compressed representation back to the original C dimensional space. This step produces C channel-wise weights, enabling element-wise recalibration with the original feature map. Without this expansion, the network would not be able to assign distinct importance weights to individual feature channels. Second, this design improves computational efficiency. A direct cross-channel mapping requires $\mathcal{O}(C^2)$ parameters, whereas the bottleneck

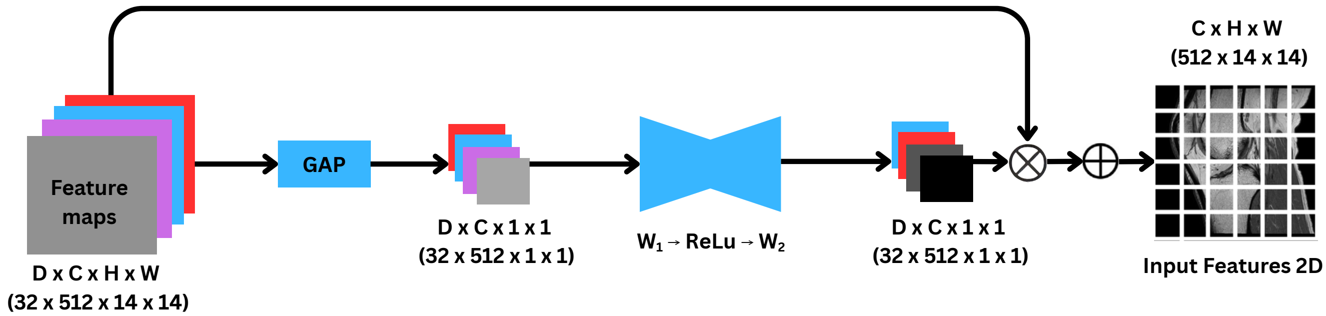


Fig. 2: **Detailed architecture of the Depth Attention module.** First, the input feature volume $\mathcal{X}$ $(D \times C \times H \times W)$ is processed by a Spatial Global Average Pooling (GAP) layer to compress the spatial dimensions, generating a global descriptor G $(D \times C \times 1 \times 1)$. Next, this descriptor is passed through a 3D convolutional bottleneck and a Sigmoid activation to compute the depth-aware attention scores $\mathcal{A} \in (0, 1)$. These scores dynamically recalibrate the original input volume via element-wise multiplication $(\otimes)$, highlighting the most informative slices containing pathological evidence. Finally, a depth-wise summation $(\sum)$ aggregates the reweighted volume into a compact 2D feature map F_{2D} $(C \times H \times W)$, preserving critical lesion features while filtering out irrelevant background noise.

reduces this to $\mathcal{O}(\frac{2C^2}{r})$. This reduction in learnable parameters acts as a form of structural regularization, limiting the model capacity and helping to mitigate overfitting on medical datasets with limited samples. Then a Sigmoid function (σ) is applied to produce an attention score $\mathcal{A} \in (0, 1)^{C \times D \times 1 \times 1}$:

$$\mathcal{A} = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 G))$$

The resulting $\mathcal{A}$ acts as a dynamic weighting mechanism, assigning higher scores to channels and slices that contain pathological evidence while reducing responses from less informative regions. The attention score $\mathcal{A}$ is then used to recalibrate the original feature volume $\mathcal{X}$ by element-wise multiplication. The reweighted features are subsequently aggregated along the depth dimension (D) to produce a feature map $F_{2D} \in R^{C \times H \times W}$:

$$F_{2D} = \sum_{d=1}^{D} (\mathcal{A}_{:,d,:,:} \otimes \mathcal{X}_{:,d,:,:})$$

Unlike standard 3D average pooling, which treats all slices equally and may dilute subtle anomalies with background noise, this depth-wise weighted aggregation emphasizes informative slices. As a result, slices with higher attention scores contribute more to the final representation, leading to a compact 2D feature map that preserves relevant information for subsequent classification.

3.3. Task-Aware Memory Mechanism

Standard classifiers often struggle to distinguish overlapping pathologies (e.g., a co-occurring ACL and meniscus tear). To address this, we propose a memory-augmented mechanism that decouples feature extraction for each specific task (e.g., separating ACL features from Meniscus features) [8]. As illustrated in Fig.1, this module operates in three steps:

3.3.1. Task-Specific Spatial Masking

As the first step in this decoupling process, the unified 2D feature map $F_{2D} \in R^{C \times H \times W}$ (where $C = 512, H = 14, W = 14$) passes through a Shared Spatial Encoder (Ψ). Structurally, this block consists of a 3×3 convolution followed by Batch Normalization and a ReLU activation [12] to generate a compressed shared feature map $F_{shared} \in R^{64 \times 14 \times 14}$:

$$F_{shared} = ReLU(BN(Conv_{3\times3}(F_{2D})))$$

By reducing the channel dimension from 512 to 64, this encoder significantly reduces computational cost while learning common structural features in all knee pathologies.

The shared representation F_{shared} is fed into separate task-specific heads. Each head acts as a spatial attention module and consists of a 1×1 convolution $(Conv^t_{1\times1})$ followed by a Sigmoid activation (σ_{sig}). This operation generates a continuous spatial probability mask $S_t \in R^{1 \times 14 \times 14}$ designed for each diagnostic task $t \in \{\text{ACL}, \text{Meniscus}\}$:

$$S_t = \sigma_{sig}\left(Conv^t_{1\times1}(F_{shared})\right)$$

The purpose of this mask S_t is to highlight the regions that are most relevant to each diagnostic task. Instead of treating the entire knee joint equally, S_t assigns spatial importance to different anatomical regions based on their relevance to a specific injury.

To recover the fine-grained pathological details lost during the 64-channel compression stage, we utilize a skip connection. The spatial attention mask generated S_t is applied to the original high-dimensional (512-channel) feature map F_{2D} by spatial broadcasting and element-wise multiplication:

$$\tilde{Z}_t = F_{2D} \odot S_t$$

By filtering out irrelevant activations, this multiplication emphasizes anatomical regions relevant to task t while reducing responses from less informative areas. The resulting masked feature map $\tilde{Z}_t \in R^{512\times14\times14}$ serves as a task-specific query that helps prevent interference between co-occurring diseases before the memory retrieval stage.

3.3.2. *Prototype Retrieval (Knowledge Referencing)*

To augment visual features with clinical prior knowledge, we utilize a Key-Value Memory Bank M_t for each task. The memory bank functions as a set of learnable diagnostic prototypes. Each memory slot is designed to capture recurring patterns, such as variations in ligament tears, meniscus lesions, or normal anatomical structures, learned in training data. Rather than relying solely on features extracted from a single input slice, the network queries the memory bank to retrieve relevant prototype representations, providing additional context for the current input. Memory size is determined by two hyperparameters: the number of memory slots N and the feature dimension D_{mem}. In our implementation, we set $N = 64$ to ensure sufficient capacity and $D_{mem} = 128$ (matching the dimension of the backbone feature) to allow direct integration. Unlike static databases, this bank is a learnable parameter matrix updated via backpropagation.

The retrieval process treats the refined feature $\tilde{Z}_t$ as the source of the Query. However, a direct dot-product attention would require the dimension of the memory feature (D_{mem}) to match 512, leading to an increased computational cost. To reduce computational cost and parameter count, we introduce a bottleneck design. A 1×1 convolution is applied to $\tilde{Z}_t$ before query generation. This operation acts as a linear projection in the channel, compressing the features of each spatial location from 512 channels to a compact embedding space with $D_{mem} = 128$, which aligns the feature dimension with that of the memory bank. Following this projection, we flatten the spatial dimensions to preserve localized information, resulting in a sequence of query vectors $q_t \in R^{(14\times14)\times128}$.We then compute their relevance to the memory prototypes using dot-product attention:[13]:

$$Score(q_t, M_t) = \frac{q_t M_t^\top}{\sqrt{D_{mem}}} \tag{1}$$

where $M_t^\top$ denotes the transpose of the memory matrix. This operation produces a similarity matrix $Score \in R^{(14\times14)\times64}$ To reduce the influence of irrelevant memory patterns, we adopt a sparse retrieval mechanism instead of applying Softmax over all memory slots. we apply a row-wise Top-K masking operation ($K = 5$) directly on the $Score$ matrix. For each spatial query in q_t, we retain only its K highest similarity scores, which correspond to the most relevant memory slots in M_t. The scores of the remaining unselected slots are forcefully set to $-\infty$. The resulting sparse scores are then normalized using a Softmax function to produce attention weights $\beta \in [0, 1]^{(14\times14)\times64}$. The final context retrieved R is then computed as the weighted sum of all memory slots utilizing these normalized scores:

$$R = \beta M_t$$

This Top-K constraint enables the model to form a focused representation by combining the most relevant memory patterns, reducing the influence of less informative memory slots. The retrieved representation R encodes contextual information from the memory bank for each spatial location. By multiplying the sparse weights β with the memory M_t the model selectively combines the feature vectors relevant prototypes of the top-K, providing contextual features to complement the original visual query.

Memory slots are continuously updated during training. At each iteration τ, the memory matrix is updated by gradient descent:

$$M_t^{(\tau+1)} \leftarrow M_t^{(\tau)} - \eta \cdot \frac{\partial \mathcal{L}_{total}}{\partial M_t^{(\tau)}} \tag{2}$$

where η denotes the learning rate and $\mathcal{L}_{total}$ is the Binary Cross-Entropy Loss.

3.3.3. *Adaptive Gated Fusion*

A static fusion of visual and memory features is suboptimal. To enable selective integration of contextual information, we introduce a dynamic context gating mechanism. The confidence gate $\lambda_t \in (0, 1)$ is computed from the concatenation of the localized visual query q_t and the retrieved context R. This combined representation is passed through a two-layer MLP with a ReLU activation, followed by a Sigmoid function:

$$\lambda_t = \sigma(\text{MLP}_{gate}([q_t \parallel R])) \tag{3}$$

where σ denotes the Sigmoid activation, $\parallel$ represents the concatenation of features. The final diagnostic representation h_t is then synthesized as an element-wise convex combination of the current query and the historical patterns retrieved, where $\odot$ denotes element-wise multiplication:

$$h_t = \lambda_t \odot R + (1 - \lambda_t) \odot q_t \tag{4}$$

This adaptive gating mechanism improves interpretability by allowing the network to learn how to balance prior memory R and visual evidence q_t. When visual features are ambiguous or affected by noise, the gate λ_t can assign a greater weight to the retrieved prototypes, providing complementary contextual information and helping to reduce uncertainty.

3.4. Multi-Label Classification Prediction

After memory retrieval, we obtain a set of disentangled feature vectors $H = \{H_{acl}, H_{men}\}$, where each $H_k \in R^{512}$ represents the refined representation for a specific diagnostic task. To capture potential correlations between pathologies (e.g., an ACL tear often accompanies a Meniscal tear), we concatenate these task-specific features into a unified diagnostic vector. This unified vector is passed through a multi-label classifier $\mathcal{C}$, composed of a fully connected layer reducing dimensions to 128, a ReLU activation [12], Dropout for regularization, giving the final prediction probabilities for ACL and Meniscus tear.

4. EXPERIMENT

4.1. Dataset

We evaluate our proposed method on the MRNet [1] and KneeMRI [14] datasets, both of which are publicly available benchmarks for automated knee MRI interpretation. The MRNet dataset [1] comprises 1,370 clinical knee MRI examinations. To ensure a fair and standardized evaluation, we report results on the official validation set.

The KneeMRI dataset [14] consists of 917 sagittal proton-density-weighted (PD-weighted) MRI series. Clinical radiologists originally annotated this dataset specifically for Anterior Cruciate Ligament (ACL) conditions across three distinct categories: intact, partially torn, and completely ruptured. However, to formulate a standard binary classification task and align with the MRNet objective, we merged the partially torn' and completely ruptured' cases into a single injured ACL' class, effectively distinguishing them from the intact' baseline.

Table 1: Statistics of the datasets used in our experiments. We report the final evaluation results on the validation sets for both MRNet and KneeMRI.

Attribute	**MRNet** [1]		**KneeMRI** [14]	
	Train	**Val**	**Train**	**Val**
Exams	1,130	120	641	276
Patients	1,088	111	641	276
Source	Stanford Univ., USA		Rijeka Hosp., Croatia	
Female (%)	530 (44.2%)		N/A	
Age (Mean)	38.1 ± 16.9		N/A	
Classes	2		2	
Scanner	GE Healthcare		Siemens Avanto	
Mag. Field	1.5 T & 3.0 T		1.5 T	

4.2. Implementation Details

Our framework is implemented using PyTorch and trained on two NVIDIA P100 GPUs. The input MRI slices are resized to 224×224 pixels and normalized. To mitigate overfitting, we apply extensive data augmentation, including random rotation ($\pm 25°$), horizontal flipping, and affine translation.

To prevent overfitting on the relatively small medical datasets and minimize computational overhead, the pre-trained ResNet-18 [9] backbone is completely frozen during training. Only the task-aware memory module, the projection heads, and the final multi-label classifier comprising exactly 650,753 trainable parameters are optimized. The model is trained for **250** epochs with a batch size of **24**. We utilize the AdamW optimizer [15] combined with a OneCycleLR scheduler, applying a peak learning rate of 1×10^{-4} exclusively to the trainable components. Training uses Automatic Mixed Precision (AMP) to optimize computational efficiency.

Pathology	**Method**	**AUC**	**ACC**	**SE**	**SP**
ACL	MRNet [1]	0.824	0.754	0.827	0.826
	David Azcona et al [16]	0.894	0.782	0.804	0.651
	Inception V4 [17]	0.851	0.651	0.622	0.708
	ELNet [18]	0.770	0.725	0.788	0.796
	Young Seok Jeon et al [19]	0.837	0.625	0.630	0.924
	MVGNN [20]	0.883	0.808	0.848	0.759
	Ali Can Kara et al [21]	0.895	0.788	0.5741	0.9688
	Ours	**0.951**	**0.9017**	**0.852**	**0.944**
Meniscus	MRNet [1]	0.745	0.620	0.550	0.700
	David Azcona et al [16]	0.821	0.782	0.837	0.742
	ELNet [18]	0.771	0.739	**0.942**	0.352
	Young Seok Jeon et al [19]	0.759	0.733	0.712	0.750
	MVGNN [20]	0.802	0.790	0.653	0.758
	Ali Can Kara et al [21]	0.798	0.771	0.569	0.625
	Ours	**0.838**	**0.805**	0.846	**0.765**

Table 2: Comparison of performance on the MRNet dataset. All methods report results based on the official MRNet validation set using single-view sagittal inputs.

4.3. Results

Table 2 summarizes the quantitative performance of our proposed method compared to various state-of-the-art (SOTA) approaches on the MRNet [1] dataset. The evaluation is conducted in the Sagittal view for both ACL and Meniscus tear detection tasks.

In the ACL detection task, our proposed method achieved robust performance (AUC **0.951**, Accuracy **0.9017**). Notably, despite being a single-view approach, our model significantly outperformed established baselines like ELNet [18] and Inception V4 [17], as well as complex multi-view systems including MRNet [1] and David Azcona et al. [16] (AUC 0.934). While the multi-view MVGNN [20] achieved a slightly higher AUC, our model demonstrated superior reliability with a Specificity of **0.944**, minimizing false positives more effectively than most competitors.

For Meniscus tear diagnosis, our method secured the highest AUC (**0.838**) and Accuracy (**0.805**), surpassing even multi-view approaches like MVGNN [20] and David Azcona et al [16]. Compared to other single-view architectures such as Young Seok Jeon et al. [19] and ELNet [18] (which suffered from a drastically low Specificity of 0.352), our model maintained the best balance between Sensitivity (0.846) and Specificity (**0.765**). These results confirm that our computationally efficient single-view framework can learn discriminative features superior to heavier multi-view ensembles.

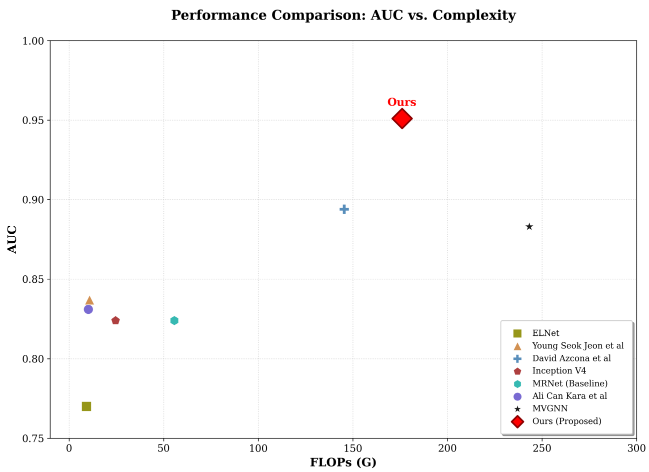


Fig. 3: Performance comparison of various models on the Sagittal view.

Figure 3 illustrates the performance comparison in terms of AUC versus computational complexity (FLOPs) among various models on the sagittal view. As shown, our proposed method achieves the highest diagnostic performance, yielding an optimal trade-off. Specifically, our model reaches a state-of-the-art AUC of 0.951 for ACL tear detection, significantly outperforming existing approaches, as previously detailed in Table 2.

To thoroughly evaluate the computational efficiency, we compare the floating-point operations (FLOPs) of our architecture against prior works (summarized in Table 3). Lightweight models such as ELNet and the architecture by Young Seok Jeon et al. [19] require only 9.21 G and 10.84 G FLOPs, respectively, but yield sub-optimal AUC scores. Standard architectures like Inception V4 [17] and the original MRNet [1] baseline operate at 24.51 G and 55.58 G FLOPs. Among the more complex diagnostic frameworks, the model by Ali Can Kara et al.[21] consumes 165.41 G FLOPs, while MVGNN [20] is the most computationally expensive, requiring 243.20 G FLOPs. In contrast, our proposed memory-augmented network requires 175.99 G FLOPs. While it introduces a moderate computational overhead compared to the baseline, it is notably more efficient than MVGNN [20] and provides a substantial performance margin over all counterparts, completely justifying the architectural design.

We also visualize the diagnostic efficacy of our proposed model through Receiver Operating Characteristic (ROC) curves for both ACL and meniscus tear detection tasks in the figure 4. The curve for the ACL task (depicted in green) exhibits an outstanding trajectory, closely hugging the upper-left corner of the plot, which culminates in a state-of-the-art Area Under the Curve (AUC) of 0.951. This indicates an exceptional balance between sensitivity (True Positive Rate) and specificity (1 - False Positive Rate), demonstrating the model's robustness in correctly identifying ACL ruptures while strictly minimizing false alarms. Furthermore, the model demonstrates commendable performance on the inherently more challenging meniscus tear task (depicted in blue), achieving a solid AUC of 0.832. The clear separation of both curves from the random-guess baseline (the diagonal dashed line) provides compelling quantitative evidence that our memory-augmented architecture successfully captures discriminative pathological features, highlighting its potential reliability for automated clinical MRI analysis.

Method	**FLOPs (G)**
ELNet [18]	9.21
Young Seok Jeon et al. [19]	10.84
Inception V4 [17]	24.51
MRNet (Baseline) [1]	55.58
Ali Can Kara et al. [21]	165.41
MVGNN [20]	243.20
Ours (Proposed)	**175.99**

Table 3: Comparison of computational complexity in terms of GFLOPs for different models on the ACL task.

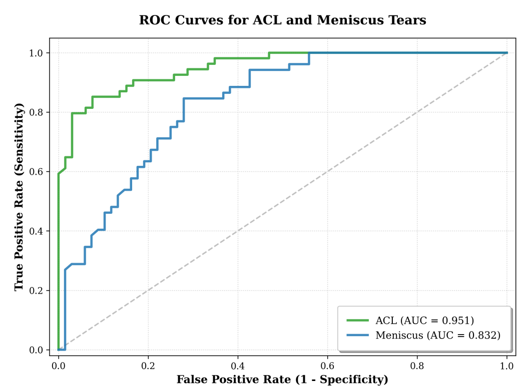


Fig. 4: Receiver Operating Characteristic (ROC) curves for the diagnostic tasks. Our proposed model achieves high sensitivity and specificity, particularly for the ACL tear detection task with an AUC of 0.951.

4.4. Ablation Study

To comprehensively evaluate the individual contribution and effectiveness of our proposed components, we conducted a systematic ablation study. All configurations are evaluated on the validation set, and the quantitative results are summarized in Table 4.

Baseline Setup: Our baseline model employs a standard ResNet-18 backbone pre-trained on Radi-Imagenet [10]. To

Table 4: Ablation study on the individual contribution of each proposed module.

Model Configuration	Average AUC
Baseline (ResNet-18)	0.850
ResNet-18 [9] + Multi-scale	0.865
ResNet-18 [9] + CBAM [7]	0.872
ResNet-18 [9] + Depth Attention	0.888
ResNet-18 [9] + Spatial Mask	0.895
Ours (SingleViewMemoryNet)	**0.925**

handle the 3D nature of the MRI volumes, this baseline utilizes a straightforward 3D Global Average Pooling layer to aggregate spatial and depth dimensions before the final linear classifier. This vanilla architecture achieves an average AUC of 0.850.

Effectiveness of Individual Modules: We independently integrate each proposed technique into the baseline to demonstrate its impact. First, incorporating the Multi-scale processing strategy improves the AUC to 0.865, demonstrating the benefit of capturing multi-resolution contextual information for localized lesions. Alternatively, integrating the Convolutional Block Attention Module (CBAM) [7] into the 2D feature extractor brings the performance to 0.872.

Notably, replacing the standard 3D average pooling with our proposed Channel-Aware Depth Attention module yields a significant boost, achieving an AUC of 0.888. This substantial improvement confirms our hypothesis: explicitly modeling cross-slice dependencies and dynamically filtering out irrelevant, healthy slices is far superior to rigid, equal-weight pooling.

Impact of the Task-Aware Memory Module: Finally, we evaluate the impact of task-specific feature refinement. Introducing the dual-head spatial mask without the memory bank improves the baseline to 0.895. However, our full pipeline—SingleViewMemoryNet—which seamlessly integrates the hybrid backbone with the complete Task-Aware Memory Module, achieves the highest average AUC of 0.925. The external memory retrieval mechanism successfully isolates, recalls, and refines the distinct diagnostic signals for ACL and Meniscus tears. This progression clearly proves the synergistic effect of our joint architecture, where each module optimally complements the others.

Table 5 shows the inference time per MRI scan for different models. Simple models like ELNet and the original MRNet are very fast (5.98 ms and 9.41 ms), but their speed often comes with lower accuracy. Our proposed model takes 32.24 ms to process one MRI scan. Although it is slightly slower due to the memory and attention components, this speed is still very fast and suitable for real-time clinical use. In comparison, more complex models like MVGNN are much slower, requiring up to 69.07 seconds per scan, which is not practical in real-world settings. Our model provides a good balance between speed and accuracy, which achieves high diagnostic performance while still maintaining a fast inference time that fits well into normal clinical workflows.

Table 5: Comparison of Inference Time among different models.

Model	Inference Time (ms)
ELNet [18]	5.98
Young Seok Jeon et al. [19]	28.25
David Azcona et al. [16]	60.57
Inception V4 [17]	17.38
MRNet (Baseline) [1]	9.41
Ali Can Kara et al. [21]	13.12
MVGNN [20]	**69.07 s**
Ours (Proposed)	**32.24**

Table 6: Comparison of model parameters on the MRNet dataset.

Method	Total Parameters	Trainable Parameters
Inception V4 [17]	41,080,000	41,080,000
Ali Can Kara et al [21]	24,702,593	1,114,881
MVGNN [20]	12,490,000	790,000
David Azcona et al [16]	11,180,000	11,180,000
MRNet (Baseline) [1]	2,470,000	2,470,000
ELNet [18]	130,000	130,000
Young Seok Jeon et al [19]	130,000	130,000
Ours	**11,827,625**	**650,753**

Table 6 compares the model complexity on the MRNet dataset [1] in terms of total and trainable parameters. Our model has 11,827,625 parameters, which puts it in a moderate range. It is much smaller than large models like Inception V4 [17] (over 41 million parameters) and the model by Ali Can Kara et al. [21] (around 24.7 million), while still being more expressive than lightweight models such as ELNet [18] (130,000 parameters) and the original MRNet [1]. Although the total number of parameters is about 11.8 million, only 650,753 parameters are actually trained due to freezing the ResNet-18 [9] during the training process. This is lower than some similar models, such as MVGNN [20] (790,000 trainable parameters) and Ali Can Kara et al. [21] (1,114,881 trainable parameters). In contrast, models like Inception V4 or the approach by David Azcona et al. [16] train all parameters in the network. This shows that our model makes effective use of transfer learning by freezing most of the parameters. As a result, it reduces training cost and helps avoid overfitting, while still keeping enough capacity to learn important patterns in medical images.

4.5. Discussion

Method	Dataset	View / Pathology	Sens.	Spec.	AUC
Our Pipeline	MRNet	Axial - ACL	**0.9259**	0.8939	**0.9604**
		Axial - Meniscus	0.8077	0.7647	0.8425
		Coronal - ACL	0.9159	**0.9537**	0.9537
		Coronal - Meniscus	0.9038	0.9091	**0.8618**
Our Pipeline	KneeMRI [14]	Sagittal - ACL	**0.8326**	**0.9219**	**0.9226**
ELNet [18]	KneeMRI [14]	Sagittal - ACL	0.7630	0.7286	0.7783

Table 7: Ablation study on different anatomical planes (Axial/Coronal) and generalization performance on the external KneeMRI dataset compared to ELNet.

To validate the generalization capability of our framework, we extended the evaluation to the additional KneeMRI dataset [14]. As detailed in Table 7, our method achieved an impressive AUC of **0.9226** for ACL detection, outperforming the ELNet [18] (AUC 0.7783) by a significant margin. This result highlights a critical advantage of our architecture: with approximately 0.65M parameters, our model strikes an optimal balance between capacity and efficiency. Unlike heavy baselines like MRNet [1] (2.5M), or lightweight models like ELNet [18] (0.21M) which may lack the representational power to generalize to unseen distributions, our method effectively captures robust features across diverse datasets.

Furthermore, we evaluated the model's adaptability on the Axial and Coronal planes of the MRNet dataset [1]. The framework demonstrated consistent effectiveness across views, notably achieving an AUC of **0.9604** and **0.8425** on the Axial plane for ACL and Meniscus tear with **0.9537** and **0.8618** on Coronal plane. It is worth emphasizing that these state-of-the-art results are achieved using a Single-View architecture. Given that different MRI planes provide complementary diagnostic information, extending this efficient pipeline into a Multi-View fusion framework could theoretically yield even superior performance, marking a promising direction for future research.

Table 8: Quantitative impact of different pre-training strategies on the proposed.

Pre-training Weights	Validation AUC		
	ACL Tear	Meniscus Tear	Average
ImageNet (Natural Images)	0.712	0.675	0.693
RadImageNet (Medical) [10]	**0.795**	**0.748**	**0.771**

Table 8 presents a quantitative comparison of different pre-training strategies and their impact on the pipeline. The results show a clear performance gap between models initialized with domain-specific medical data and those pre-trained on natural images. By using weights pre-trained on RadImageNet [10] leads to a noticeable improvement in overall performance, with the average Validation AUC increasing from 0.693 to 0.771. This trend is consistent across both primary tasks. For ACL Tear detection, the AUC improves from 0.712 to 0.795, while for Meniscus Tear detection, it increases from 0.675 to 0.748. These results suggest that pre-training on medical imaging data provides more relevant feature representations, which in turn enhances performance on MRI-based tasks. Overall, the findings highlight the importance of domain-specific pre-training for specialized medical image analysis.

Task	Pre-training Weights	AUC	ACC	SE	SP
ACL	ImageNet-1K [22]	0.901	0.808	0.813	0.833
	RadImageNet [10]	**0.951**	**0.902**	**0.852**	**0.944**
Meniscus	ImageNet-1K [22]	0.809	0.791	0.740	0.754
	RadImageNet [10]	**0.838**	**0.805**	**0.846**	**0.765**

Table 9: Ablation study evaluating the impact of pre-training strategies on our proposed framework. The model demonstrates robust performance even when utilizing standard ImageNet weights, proving the inherent efficacy of the architectural design.

To examine the effect of pre-training and address potential bias, we conduct an ablation study comparing ImageNet-1K [22] and RadImageNet [10] initializations (Table 9). As expected, RadImageNet [10] achieves the best performance (AUC 0.951 for ACL and 0.838 for Meniscus), benefiting from domain-specific medical knowledge. However, even with standard ImageNet-1K [22] pre-training, our model still performs strongly, achieving an AUC of 0.901 for ACL and 0.809 for Meniscus. By only using the pre-train Resnet-18 [9] in ImageNet-1K [22], this version already outperforms most existing methods in Table 2. For example, in ACL detection, it surpasses the strongest baseline (Ali Can Kara et al.,[21] AUC 0.895). For Meniscus detection, it remains competitive and exceeds methods such as MVGNN [20] (AUC 0.802) and ELNet [18] (AUC 0.771). These results indicate that the performance gain of our framework does not rely solely on advanced pre-training, but mainly comes from the effectiveness of the proposed memory module and spatial aggregation design.

Figure 5 shows the refined "Visual Dictionary" with five memory slots (Slots 2, 11, 30, 47, and 59) that are highly related to ACL tear detection. Each slot captures a specific visual pattern, suggesting that the model learns to break down the pathology into smaller, meaningful components. Slots 2, 11, 30, and 59 mainly focus on edema-related patterns. They respond strongly to bright (hyperintense) regions in the MRI, which usually indicate fluid accumulation, joint effusion, or hemorrhage. These are common signs associated with ACL injuries, especially around the intercondylar notch and bone attachment areas. In contrast, Slot 47 focuses more on structural changes of the ligament. Instead of detecting edema, it

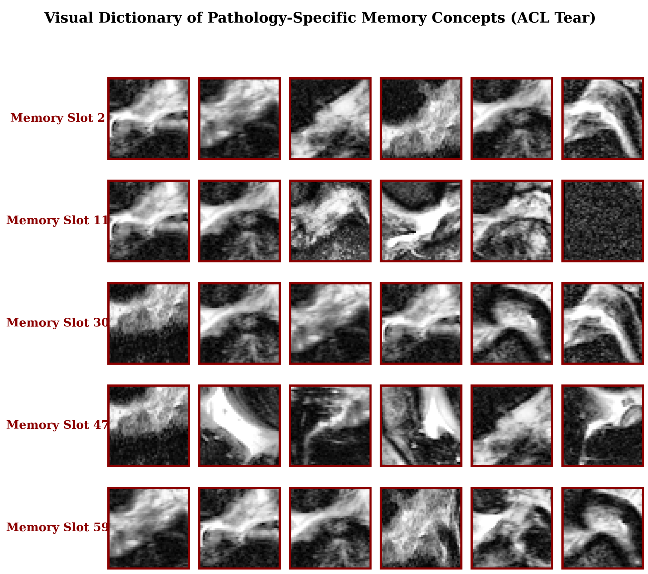
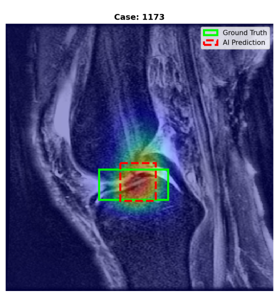
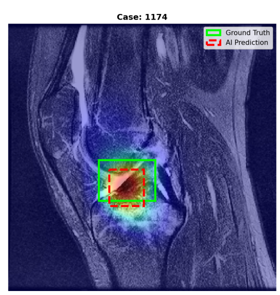


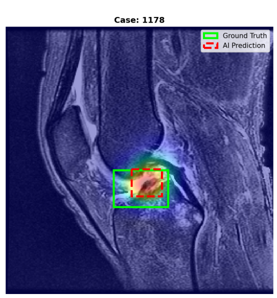
Fig. 5: Visual dictionary showcasing top activating patches across five selected pathology-specific memory slots (Slots 2, 11, 30, 47, 59) for ACL tear detection.



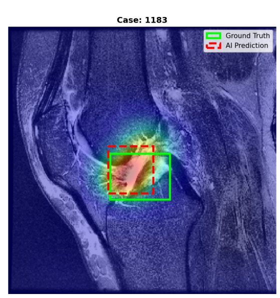
(a) Case 1173 (IoU = 0.6)



(b) Case 1174 (IoU = 0.7)



(c) Case 1178 (IoU = 0.6)



(d) Case 1183 (IoU = 0.8)

Fig. 6: The green solid boxes denote the physician-annotated ground truths, while the red dashed boxes represent our model's predictions extracted from the memory slot's attention heatmaps. The visual overlap demonstrates the model's precise capability to localize lesion areas.

captures abnormal shapes, such as broken or irregular fibers, and the appearance of ligament remnants after a complete tear. The consistent high activation (red borders) shows that these slots are mainly triggered by abnormal regions rather than normal anatomy. This suggests that the model is not simply acting as a black box, but is learning meaningful visual patterns that are relevant for diagnosis, making its predictions easier to interpret.

To evaluate the robustness and generalizability of our method, we conduct cross-dataset experiments between the KneeMRI [14] and MRNet datasets (Table 10). The results show that the model maintains strong performance across different imaging settings. When trained on KneeMRI [14] and tested on MRNet [1], the model achieves a high sensitivity of 0.9639 and achives an AUC of 0.8460, indicating that it can correctly detect most ACL tear cases. In the reverse setting, training on MRNet [1] and testing on KneeMRI [14], the model achieves an AUC of 0.8614 and a specificity of 0.8950, showing good overall performance while keeping false positives low. These results suggest that the model generalizes well across datasets with different acquisition conditions. In particular, the Task-Aware Memory Module helps capture consistent anatomical features, making the model more robust to distribution differences in real-world clinical data.

Table 10: Cross-Dataset Generalization Results for ACL Tear Detection

Experiment Setting	AUC	Accuracy	Sensitivity	Specificity
KneeMRI $\rightarrow$ MRNet	0.8406	0.7833	**0.9639**	0.7364
MRNet $\rightarrow$ KneeMRI	**0.8614**	**0.7941**	0.7909	**0.8950**

We use Grad-CAM [23] to generate heatmaps highlighting the regions that contribute most to the model's predictions, confirm that our framework not only achieves high AUC performance but also learns clinically meaningful representations by precisely localizing the damaged regions of the ACL and meniscus, rather than relying on false correlations. As shown in Fig. 7, the model accurately localizes ACL and meniscus tears in the Sagittal view. The highlighted regions align well with clinically relevant features, such as ligament discontinuity and localized edema. To further assess robustness, we also evaluate the model on Axial and Coronal views (Fig. 8). The attention maps remain consistent across these different views, indicating that the model captures meaningful anatomical features rather than relying on dataset-specific patterns. Overall, this consistent behavior across views suggests that the model has strong spatial awareness and can support multi-view clinical analysis similar to how radiologists examine MRI scans.

To evaluate the spatial accuracy of our framework, we use the Intersection over Union (IoU) metric. IoU measures the overlap between the predicted bounding box (B_{pred}), obtained by thresholding the task-aware memory attention heatmaps, and the ground truth bounding box (B_{gt}) annotated by physicians. It is defined as:

$$\mathrm{IoU} = \frac{|B_{pred} \cap B_{gt}|}{|B_{pred} \cup B_{gt}|} \tag{5}$$

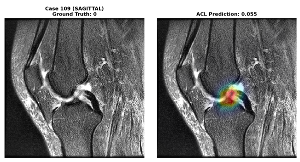
Figure 6 demonstrates the localization results on several validation cases (1173, 1174, 1178, and 1183). The predicted regions (red dashed lines) show strong overlap with the ground truth annotations (green solid lines), indicating good IoU alignment. This suggests that the model focuses on relevant pathological features, such as fluid accumulation and ligament damage, rather than background regions. This localization performance is achieved under a weakly supervised setting, using only image-level labels without requiring bounding-box or pixel-level annotations. These results support both the interpretability and reliability of our method.

5. CONCLUSION

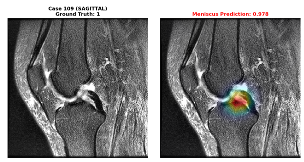
The proposed method achieves high diagnostic performance while remaining efficient and transparent. By combining anatomy-based localization with a 'task-aware memory' mechanism, the model successfully captures key diagnostic features without the high computational cost of full 3D networks. Our visual analysis confirms that the model focuses on the correct anatomical landmarks, mirroring the radiologist's reasoning process. Our framework demonstrates that it relies on valid structural evidence rather than image artifacts. Ultimately, this research offers a practical, explainable solution that is well-suited for deployment in clinical environments with limited computing resources.

Figure 7: Sagittal View

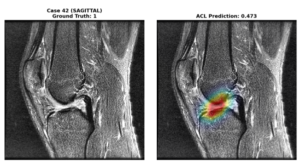
ACL Tear | **Meniscus Tear**



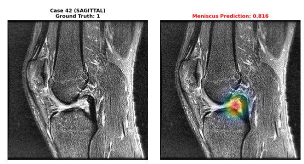
(a) Case 109 ($p = 0.95$)



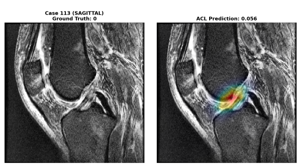
(b) Case 109 ($p = 0.94$)



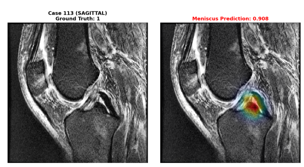
(c) Case 42 ($p = 0.97$)



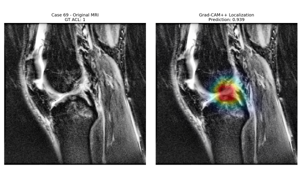
(d) Case 42 ($p = 0.96$)



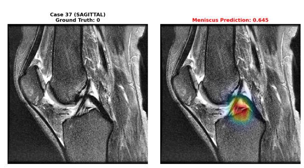
(e) Case 113 ($p = 0.95$)



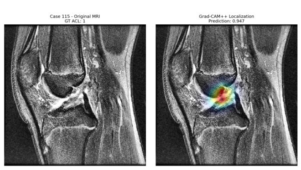
(f) Case 113 ($p = 0.87$)



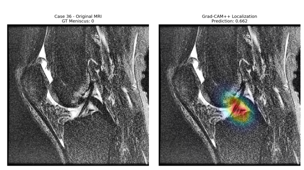
(g) Case 69 ($p = 0.94$)



(h) Case 37 ($p = 0.86$)

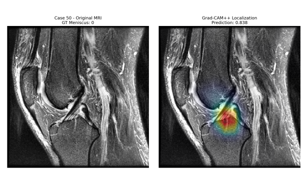


(i) Case 115 ($p = 0.95$)

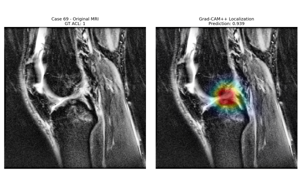


(j) Case 36 ($p = 0.90$)

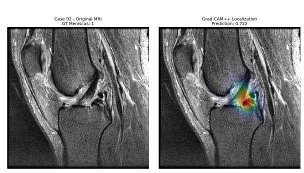
(k) Case 118 ($p = 0.95$)



(l) Case 50 ($p = 0.83$)



(m) Case 69 ($p = 0.94$)

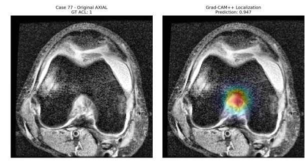


(n) Case 92 ($p = 0.82$)

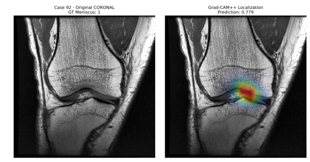
Fig. 7: GRADCAM in Sagittal view. p: predicted probability.

Figure 8: Coronal & Axial View

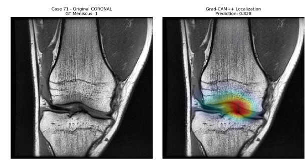
ACL Tear | **Meniscus Tear**



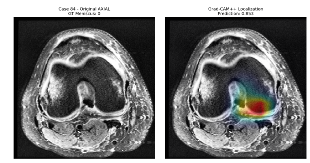
(a) Case 77, Axi ($p = 0.95$)



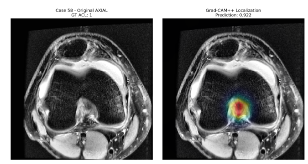
(b) Case 77, Axi ($p = 0.87$)



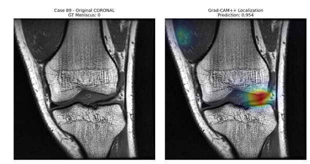
(c) Case 77, Cor ($p = 0.83$)



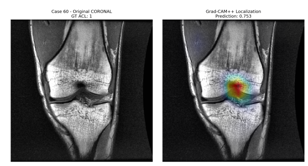
(d) Case 77, Cor ($p = 0.94$)



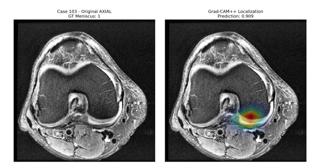
(e) Case 58, Axi ($p = 0.92$)



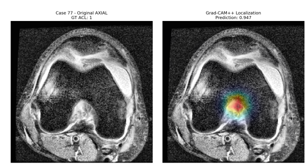
(f) Case 58, Men ($p = 0.90$)



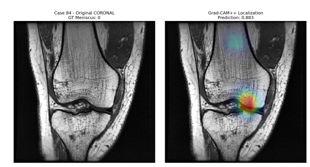
(g) Case 103, Axi ($p = 0.85$)



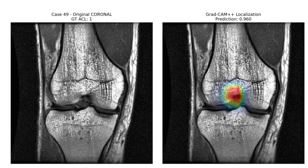
(h) Case 103, Men ($p = 0.91$)



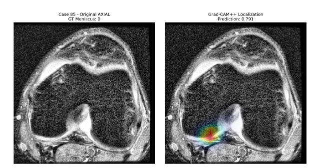
(i) Case 95, Axi ($p = 0.85$)



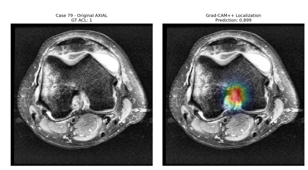
(j) Case 84, Men ($p = 0.88$)



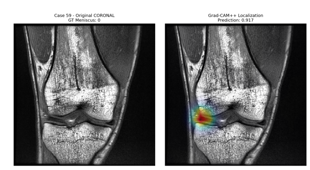
(k) Case 49, Axi ($p = 0.96$)



(l) Case 85, Men ($p = 0.89$)



(m) Case 79, Axi ($p = 0.90$)



(n) Case 59, Men ($p = 0.92$)

Fig. 8: GRADCAM in Coronal and Axial view. p: predicted probability.

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